

Unit 5: Fundamentals of Solid state Physics.

Crystal bonding:

In a crystal lattice atoms, molecules or ions are held together in particular position as a basis. The force which linked atoms in a crystal at definite lattice position is term as crystal bonding. The corresponding energy involved in crystal bonding is called crystal bond energy.

Hence, it also can be define as the amount of energy that is required to separate its constituents atoms, molecules or ions ^{to} infinite separation. ~~at neutral state.~~

The different types of bonding present in crystal are as follow,

1. Ionic bonding:

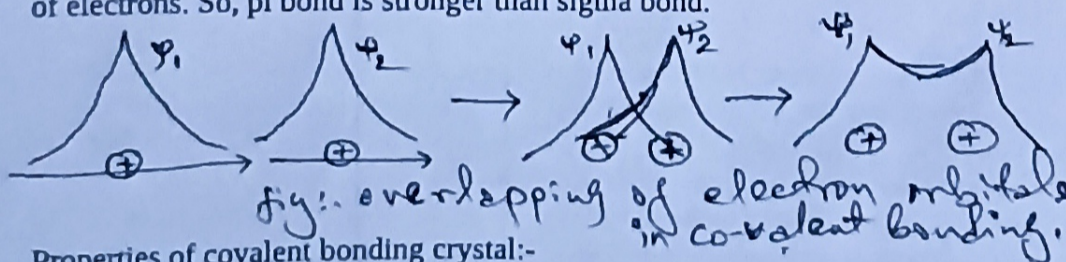
The bond developed between two atoms by complete transfer of electron i.e. either gain or loss of electron by atoms is called ionic bonding. In ionic bonding atoms are bounded together by coulomb electrostatic force. This types of bonding is present in ionic crystal for e.g. NaCl, LiF, ZnS e.t.c.

Properties of ionic bonding crystal :-

- It is strongest bond.
- Crystal with this bond are usually hard and brittle in nature.
- They are highly soluble in non polar solvent (water).
- Crystal with this bond are non-conducting in solid phase while its aqueous state is good conductor of electricity.
- They have high and sharp value of melting and boiling point.

2. Co-valent bonding:

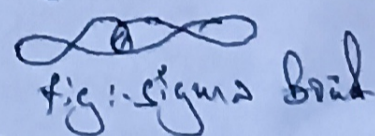
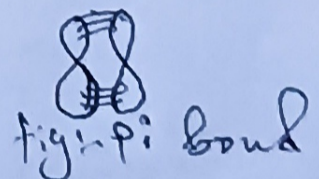
The bond developed between two atoms because of sharing of electrons to gain their nearest inertness state of atom is called co-valent bonding. The atoms usually with tetra valence electron like carbon, silicon formed this type of bonding and hence have tetrahedral structure. In co-valent bonding, electron clouds is slightly overlapping such that electron can be share to both atom equally. Diamond, Silicon carbide (SiC), H_2 gases molecules possess this type of bonding. Co-valent is further of two types pi or sigma covalent bond; pi bond consists of sidewise or parallel overlapping of orbitals of electrons where in sigma bond case it has head on overlapping of orbitals of electrons. So, pi bond is stronger than sigma bond.



Properties of covalent bonding crystal:-

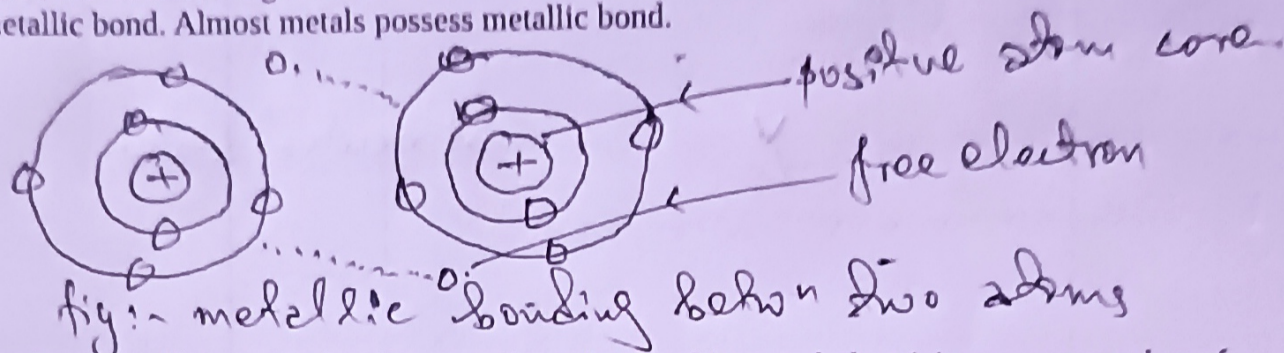
- It is strongest bond rest to that of ionic bond.
- Crystals with co-valent bonding are insoluble in non polar solvent but soluble in polar solvent (alcohol).
- Crystals are hard with higher value of melting and boiling point.
- The conductivity of crystal increases with temperature. The conductivity varies widely i.e. diamond is insulators in nature while silicons are semi conductor in nature.
- Co-valent crystals are transparent to longer wavelength radiation and opaque to shorter wave length radiation.

Note : if energy gap in crystal is larger than energy of particular photon or light then crystal are transparent to given photon or light otherwise it gets absorbed by crystal and hence crystals are opaque.



3. Metallic bonding:-

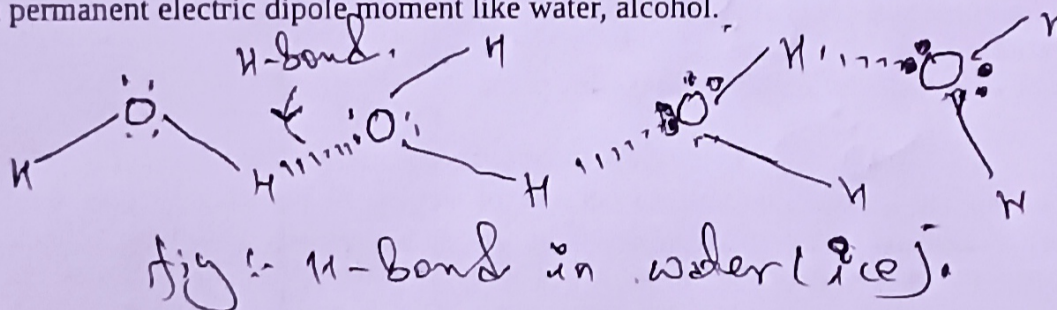
A metal consists of loosely bound valence electron at their outermost orbit and hence are free to move randomly throughout the crystal. Thus, in metallic conductor crystal +ve charge atomic core are bounded together because of the delocalized free electron such a bond in metal is called metallic bond. Almost metals possess metallic bond.



Crystal with metallic bond are good conductor of heat and electricity as a number of free electrons present in it are carries for transport of heat and electricity. These crystals have high value of electrical and thermal conductivity. Metallic crystal are also good reflector of electromagnetic radiation. Example of crystal with metallic bond are Cu, Al, Fe e.t.c.

4. Hydrogen bonding:-

The bond develop between hydrogen atom of one molecule with more electronegativity atom of another molecule is called hydrogen bonding. A hydrogen bonding is present in a molecule with permanent electric dipole moment like water, alcohol.



In hydrogen bonding molecules with more electronegative atom acts as acceptor of electron while molecules with hydrogen atom acts as donor species. Hydrogen bond is stronger than vanderwall bonding. A crystal like H_2O (ice), NH_3 , HF e.t.c. contains hydrogen bonding.

5. Vanderwall bonding:

Inert atoms with low atomic number have closed electronic shells and hence two atomic nuclei along with their orbital electrons are bound together as they approach sufficiently close to each other such bonding in atoms is called Vanderwall bonding. When two atoms are close together positive nucleus of one atom attracts orbital electrons of another towards it and hence a temporary dipole is developed and have definite dipole moment. In this way an electrostatic interaction between two temporary dipoles results to vanderwall bonding.

This is the weakest bond present in crystal. Crystal like graphite, solid argon, methane have Vanderwall bonding. This bonding is in existence when molecules are close enough only.

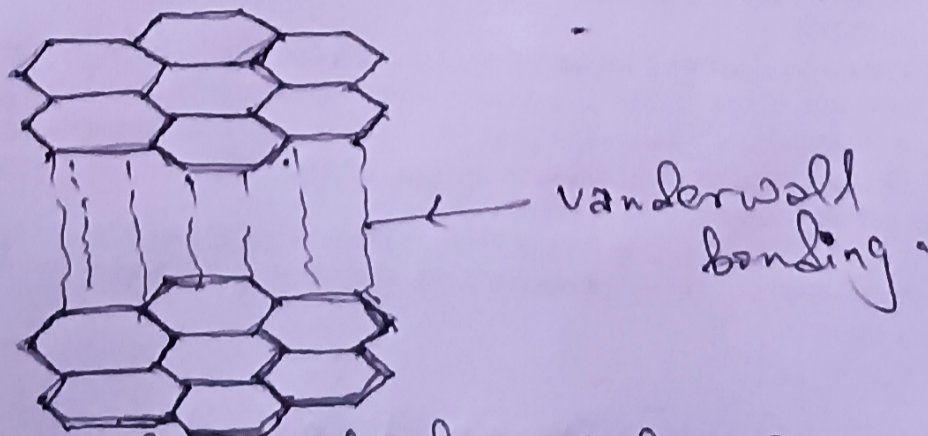


fig:- Vanderwall bonding betwⁿ two layers of graphite.

Free electron theory of metal:-

In metal, atoms are bounded together because of loosely bound orbital free electrons. These electrons are free to move within crystal boundaries just like gas enclosed in a container and hence these electrons in metal are also called as electron gas.

In free electron theory of metal, the movement of free electron inside metal is equivalent to just like non-interacting gas molecules that are enclosed in potential energy box. Electrons inside metal are non interacting and obey Pauli-exclusion principle due to which they are also named as free electron Fermi gas particles.

There is basically two empirical model in order to explain Free electron theory of metal to explain their electrical and heat conducting nature.

1. Classical Free Electron model (CFE).
2. Quantum Mechanical Free Electron model (QMFE).

1. Classical Free Electron model (CFE):-

The main assumption of Classical free electron model are as follow;

a. Metal is composed of an array of ions with free valence electrons that can move within Solid crystal boundaries and are the carriers for heat and electricity.

b. The possible repulsive interaction present between electrons inside the metal is neglected and the P.E due to atomic ion core is assumed to be constant every where such that free electron acts as non-interacting identical particles and follow Maxwell- Boltzmann statistics.

c. In absence of an external electric field, electrons inside metal are in random motion with thermal velocity 1.2×10^5 m/sec at room temperature due to thermal energy. Hence, there is no net electric conduction in metal. i.e.

K.E of an electron = thermal energy

$$\text{or, } \frac{1}{2} m v^2 = 3 \cdot \frac{1}{2} K_B T$$

At room temperature; $T = 27^\circ \text{C} = 300\text{K}$ So,

$$\text{or, } v = \sqrt{\frac{K_B T}{m}} = \sqrt{\frac{1.38 \times 10^{-23} \times 300}{9.1 \times 10^{-31}}} \\ = 1.2 \times 10^5 \text{ m/Sec}$$

d. When an external electric field is applied, free electrons in metal acquire an average drift velocity and start to move in opposite direction to applied electric field. Hence, there is electrical conduction in metal.

With all these assumptions, Classical free electron theory is successful to predict Ohm's law, Wiedemann Frantz law and explaining empirical formula related to electrical and thermal conductivity in metal.

Expression for electrical conductivity of metal from Classical free electron model (CFE):-

Let, us consider a metallic conductor having 'l' length, 'A' cross-sectional area and containing 'n' number of electron density i.e. number of electrons per unit volume. As from classical theory of free electron, electrons are moving in random direction with root mean square velocity ' $v_{r.m.s}$ ' due to definite thermal energy and hence there is no net flow of charge or electric conduction in conductor.

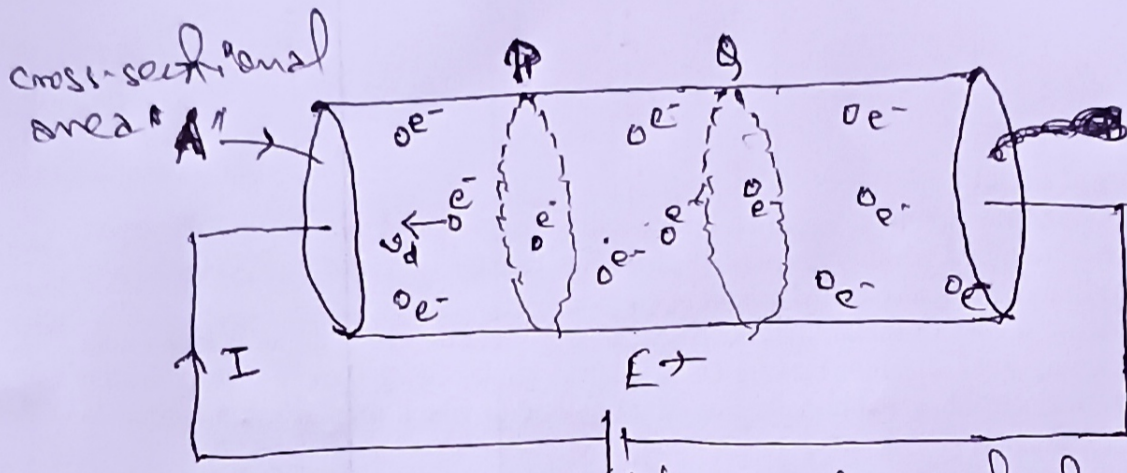


fig:- flow of charge inside conductor with V p.d. across it.

For, ' V ' be potential difference applied across end of conductor and ' E ' be resultant electric field set up inside a conductor due to p.d. Then, the coulomb force experience by an electron in ' E ' field strength is

$$F = eE \dots\dots\dots 1$$

Because of this force, electrons inside metal start to move in opposite direction to applied field and suffer a numbers of collision with +ve atomic ion cores on its way such that they have different velocities within different numbers of collision. So, an average velocity is defined for an electron inside a conductor which is called drift velocity.

For ' v_d ' be drift velocity gain by electron and ' q ' be charge passing through a given cross-section of conductor in ' t ' time. Then, current in conductor is

$$I = \frac{q}{t} \dots\dots\dots 2$$

Further, for ' p ' and ' Q ' be two cross-sectional planes at a distance of ' $v_d \cdot t$ '. Then, total charge flowing in time ' t ' through conductor be equal to total charge present within volume of cylinder PQ.

Now, total charge present in PQ cylinder region (q) = total number of free electron present in PQ cylinder X charge of electron.

$$\text{or, } q = n A v_d t \cdot (-e) \dots\dots\dots 3$$

from eqns 3 and 2 ; we get,

$$I = \frac{-n A v_d t e}{t}$$

$$\text{or, } I = -n A v_d e \dots\dots\dots 4$$

This is required expression for current flowing through metallic conductor. The -ve sign represent the drift velocity of an electron inside conductor is opposite to applied electric field direction.

$$\text{Further, current density } J = \frac{I}{A} = n e v_d \dots\dots\dots 5$$

from Newton's 2nd law of motion; for ' a ' be acceleration of an electron due to ' E ' force field, then

$$F = eE$$

$$\text{or, } m a = eE$$

$$\text{or, } a = \frac{e}{m} E$$

$$\text{or, } \frac{dv}{dt} = \frac{e}{m} E$$

$$\text{or, } dv = \frac{e}{m} E dt$$

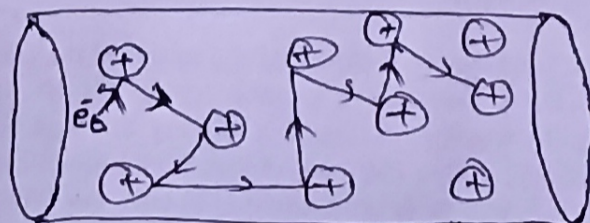


fig:- electron with drift velocity & mean free path inside conductor.

Integrating on both sides; we have,

$$\int_0^{v_d} dv = \int_0^{\tau} \frac{e}{m} E dt$$

where 'τ' is called relaxation time and is define as the time period between two successive collision of an electron with atomic core inside conductor.

$$\text{or, } v_d = \frac{e}{m} E \tau \dots\dots\dots 6$$

With this eqn 5 becomes;

$$J = ne \frac{e}{m} E \tau$$

$$\text{or, } J = \frac{ne^2 \tau}{m} E \dots\dots\dots 7$$

again, the current density in term of conductivity of metal is define as

$$J = \sigma E \dots\dots\dots 8$$

Comparing, eqns 6 and 7, we get;

$$\sigma = \frac{ne^2 \tau}{m}$$

This is required expression for electrical conductivity of metal. Hence, from this it is also proved the Ohm's law in metallic conductor.

Further, As $\tau = \frac{\lambda}{v}$; where 'λ' is mean free path of an electron and

$$v = v_{r.m.s} + v_d ; \text{ as } v_d \ll v_{r.m.s}$$

So, $v = v_{r.m.s}$

The value of $v_{r.m.s}$ doesn't depend on applied field 'E'. So, this implies that electrical conductivity 'σ' of conductor is independent of applied electric field 'E'.

Expression for electrical conductivity of metal from Quantum mechanical free electron theory:-

In a presence of electric ' $\vec{E}$ ' and magnetic field ' $\vec{B}$ ', the force experience by an moving electron is given by Lorentz force.

$$\vec{F} = -e \vec{E} + \frac{(\vec{v} \times \vec{B})(-e)}{c}$$

$$\text{or, } \vec{F} = -e \left[\vec{E} + \frac{(\vec{v} \times \vec{B})}{c} \right] \dots\dots\dots 1$$

where, $\vec{v}$ is the velocity of moving electron.

In quantum approach, momentum of particle is expressed in term of wave vector 'k' as

$$\vec{p} = \hbar \vec{k} \dots\dots\dots 2$$

Then, the force on electron from Newton's 2nd law is

$$\vec{F} = \frac{d\vec{p}}{dt} = \frac{d(\hbar \vec{k})}{dt} = \hbar \frac{d\vec{k}}{dt} \dots\dots\dots 3$$

from eqn 1 and 3;

$$\hbar \frac{d\vec{k}}{dt} = -e \left[\vec{E} + \frac{(\vec{v} \times \vec{B})}{c} \right]$$

Since in case of current carrying conductor only electric field is present inside conductor and magnetic field is not applied i.e. $\vec{B} = 0$

So,

$$\hbar \frac{d\vec{k}}{dt} = -e \vec{E}$$

$$\text{or, } d\vec{k} = \frac{-e}{\hbar} \vec{E} dt$$

An application of constant electric field displaces the Fermi sphere in k-sphere from $k(0)$ to $k(E)$;
Now, integrating we get;

$$\int_{k(0)}^{k(E)} d\vec{k} = - \int_0^t \frac{e}{\hbar} \vec{E} dt$$

$$\text{or, } \delta\vec{k} = \frac{-e}{\hbar} \vec{E} \tau \dots\dots\dots 4 \text{ where, } \delta\vec{k} = k(E) - k(0) \text{ is difference in momentum vector of electron on application of field.}$$

Further, as $p = mv = \hbar k$

$$\text{or, } k = \frac{m}{\hbar} v$$

$$\text{or, } \delta k = \frac{m}{\hbar} \delta v$$

with this equation 4 becomes;

$$\frac{m}{\hbar} \delta v = \frac{-e}{\hbar} \vec{E} \tau$$

$$\text{or, } m \delta v = -e \vec{E} \tau$$

$$\text{or, } \delta v = \frac{-e \tau}{m} \vec{E} \dots\dots\dots 5$$

Then, the electron current density in conductor is given by,

$$J = n(-e) \delta v$$

$$\text{or, } J = n(-e) \frac{-e \tau}{m} \vec{E}$$

$$\text{or, } J = \frac{n e^2 \tau}{m} \vec{E}$$

Comparing above relation with ; $J = \sigma E$

$$\sigma = \frac{n e^2 \tau}{m}$$

which is required expression of electrical conductivity. The reciprocal of electrical conductivity gives electrical resistivity as

$$\rho = \frac{1}{\sigma} = \frac{m}{n e^2 \tau}$$

Thermal Conductivity of metal from classical free electron model (CFE):-

Thermal conductivity of metal is define as the transport of heat energy due to random molecular motion across a temperature gradient. This is the measure of ability of material to conduct heat through it. It is denoted by 'k' or ' λ '.

The rate of amount of heat transfer across conductor of 'A' cross-sectional area in 'dT' temperature difference within 'dl' length is given by

$$\frac{Q'}{t} = \frac{k A dT}{dl}$$

or, $Q = \frac{k dT}{dl} \dots\dots\dots 1$ where $Q = \frac{Q'}{A \cdot t}$ = heat flux i.e. flow of heat per unit area per sec.

Mathematically, coefficient of thermal conductivity 'k' of a metal is also define as the heat flux that flows through a rod with unit temperature gradient at the steady state. i.e.

$$k = \frac{Q}{(dT/dl)} \dots\dots\dots 2$$

Let, us consider a metal that consists of 'n' number of free electron concentration which are moving with 'v' thermal velocity through 'A' cross-sectional area of metal.

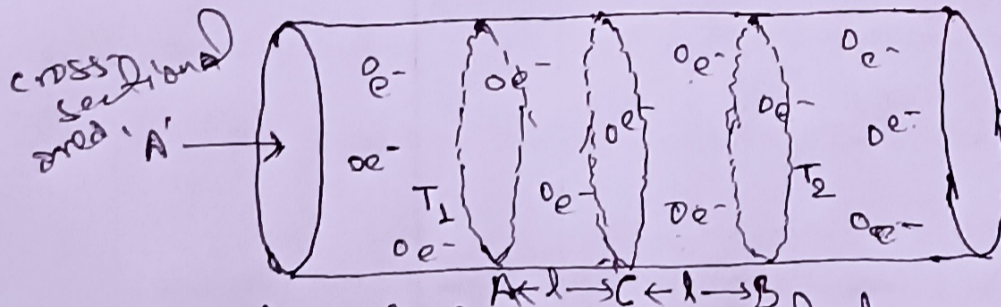


fig. electron inside conductor under thermal conduction

Further, let us consider a reference plane 'C' inside a conductor and two surfaces 'A' and 'B' such that 'A' surface is at 'T₁' higher temperature and 'B' at 'T₂' lower temperature each at 'l' distance of separation from 'C'. The distance between two surfaces 'A' and 'B' is '2l'.

The number of electrons crossing plane C to right side from left per unit area per second is

$$n' = \frac{nv}{6} \dots\dots\dots 2$$

Now, the heat energy carried by electron to right from left side crossing 'C' plane per unit its area per unit time is

$$Q_1 = n' c_e T_1$$

or, $Q_1 = \frac{nv}{6} c_e T_1 \dots\dots\dots 3$ [from eqn 2]

where, C_e is thermal heat capacity of electron per unit volume.

Simillarly, the heat energy carried to left from right side crossing 'C' plane per unit area per unit time is

$$Q_2 = n c_e T_2$$

$$\text{or, } Q_2 = \frac{nv}{6} c_e T_2 \dots\dots\dots 4$$

Now, net heat energy transfer per unit area per unit second is

$$Q = Q_1 - Q_2$$

$$\text{or, } Q = \frac{nv}{6} c_e T_1 - \frac{nv}{6} c_e T_2 \quad [\text{being } T_1 > T_2]$$

$$\text{or, } Q = \frac{nv}{6} c_e (T_1 - T_2) \dots\dots\dots 5$$

Further, The temperature gradient is expressed as;

$$\frac{dT}{dl} = \frac{T_1 - T_2}{2l}$$

$$\text{or, } T_1 - T_2 = 2l \frac{dT}{dl}$$

with this equation 5 becomes;

$$Q = \frac{nv}{6} c_e \times 2l \frac{dT}{dl}$$

$$\text{or, } Q = \frac{nv}{3} c_e l \frac{dT}{dl} \dots\dots\dots 6$$

Since eqn 1. and 6. represent the same quantity So,

$$\frac{k dT}{dl} = \frac{nv}{3} c_e l \frac{dT}{dl}$$

$$\text{or, } k = \frac{nv}{3} c_e l \dots\dots\dots 7$$

This is required expression for heat conduction coefficient from classical free electron model.

Further, from quantum approach, for a Fermi gas thermal heat capacity of electron per unit volume can be expressed in term of Fermi temperature as

$$c_e = \frac{\pi^2}{2} k_B \frac{T}{T_f} \dots\dots\dots 8$$

where Fermi energy $E_f = K_B T_f$ and $E_f = \frac{1}{2} m v_f^2$ with these relations

$$c_e = \frac{\pi^2}{2} k_B \frac{T}{\cancel{E_f} / k_B}$$

$$\text{or, } c_e = \frac{\pi^2}{2} k_B^2 \frac{T}{E_f}$$

$$\text{or, } c_e = \frac{\pi^2}{2} k_B^2 \frac{T}{\frac{1}{2} m v_f^2}$$

$$\text{or, } c_e = \pi^2 k_B^2 \frac{T}{m v_f^2} \dots\dots\dots 8$$

Then, the expression for thermal conductivity becomes; from eqn 7 and 8.

$$k = \frac{n v_f}{3} \pi^2 k_B^2 \frac{T}{m v_f^2} l \quad [\text{being } v = v_f \text{ is Fermi velocity}]$$

$$\text{or, } k = \frac{n}{3} \pi^2 k_B^2 \frac{T}{m v_f} l$$

$$\text{or, } k = \frac{n}{3} \pi^2 k_B^2 \frac{T}{m v_f} \tau v_f \quad [\text{being } l = \tau v_f \text{ is mean free path of electron}]$$

$$\text{or, } k = \frac{\pi^2 n}{3 m} k_B^2 T \tau$$

This is required modified expression for thermal conductivity of metal in term of relaxation time of free electron inside it.

Wiedemann -Franz law:-

We know, for a metal with 'n' number of electrons per unit volume.

$$\text{Electrical conductivity } \sigma = \frac{n e^2 \tau}{m} \dots\dots\dots 1 \quad \text{and}$$

$$\text{thermal conductivity } k = \frac{\pi^2 n}{3 m} k_B^2 T \tau \dots\dots\dots 2$$

Then, the ration of thermal conductivity to electrical conductivity is given by

$$\frac{k}{\sigma} = \frac{\frac{\pi^2 n}{3 m} k_B^2 T \tau}{\frac{n e^2 \tau}{m}}$$

$$\text{or, } \frac{k}{\sigma} = \frac{\pi^2}{3} \left(\frac{k_B}{e} \right)^2 T$$

$$\text{or, } \frac{k}{\sigma} = 2.45 \times 10^{-8} T \quad \text{where, } \frac{\pi^2}{3} \left(\frac{k_B}{e} \right)^2 = 2.45 \times 10^{-8} \text{ watt ohm/deg}^2 \text{ is}$$

constant term known as Lorentz unit.

$$\text{Or, } \frac{k}{\sigma} \propto T$$

This relation is known as Wiedemann- Franz law. This law states that " for a metallic conductor at not too low temperature, the ration of thermal conductivity to electrical conductivity due to free electron present in it is directly proportional to its absolute temperature."

Drawbacks of classical free electron model:-

1. According to classical free electron model, the contribution on specific heat capacity in metal by conduction electron is constant for all range of temperature i.e. C_v is independent of temperature.

$$C_v = \frac{3}{2} R \quad \text{where } R \text{ is gas constant.}$$

But, experimentally specific heat capacity per mole is proportional to temperature i.e

$$C_v = 10^{-4} R T$$

2. Classical free electron model explains that electrical conductivity of metal at low temperature is inversely proportional to square root of temperature i.e.

$$\text{electrical conductivity } \sigma = \frac{n e^2 \tau}{m} \quad \&$$

$$\text{relaxation time } \tau = \frac{\lambda}{v_{r.m.s}}$$

$$\text{or, } \tau = \frac{\lambda}{\left(\frac{3 K_B T}{m}\right)^{1/2}} \quad \left[\text{being } \frac{1}{2} m v_{r.m.s}^2 = 3 \cdot \frac{1}{2} K_B T \right]$$

$$\text{or, } \tau \propto \frac{1}{T^{1/2}}$$

$$\text{So, as } \sigma \propto \tau \propto \frac{1}{T^{1/2}} = T^{-1/2}$$

But, experimentally for a metal at low temperature the

electrical conductivity is inversely proportional to temperature i.e. $\sigma \propto \frac{1}{T}$

Quantum Mechanical Free Electron model [QMFE] :-

In order to address the failure of classical free electron model on issue of temperature dependence of metal for its specific heat capacity and electrical conductivity, Sommerfield modified the free electron model and give new approach named Quantum mechanical free electron model.

The main assumptions of quantum mechanical free electron model are:-

1. The energy level of electron gas is quantized in metal and hence electron is treated as quantum particle.
2. The electron in metal obey Pauli exclusion principle this implies no more two electrons can be in same energy state with same value of quantum numbers. Hence, quantum mechanical gas will follow Fermi- Dirac Statistical distribution.

Hence, in quantum mechanical free electron model, the valence electrons retain the main features of an ideal gas but these must be treated as quantum particles.